
Objectives:

- Transform a client request into a formal IEEE 830 Software Requirements Specification (SRS).
 - Identify and classify explicit, implicit, and missing requirements through stakeholder and context analysis.
 - Translate requirements into structured use cases and user stories for agile development.
 - Reflect on how formal analysis improves communication, estimation, and product quality in real projects.
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Task 1/ UNDERSTANDING THE CONTEXT & STAKEHOLDER ANALYSIS

Phase 1A: Initial Reading & Comprehension

Instructions: Read the client request document carefully. Answer the following questions to ensure you understand the domain and client needs.

1. Organization Context

- o Who is the client? What do they do?
- o What is the current state of their digital assets?
- o What is the primary pain point that drives this project?

2. Project Scope Understanding

- o What problem are they trying to solve?
- o Why is it important to them (business/institutional justification)?
- o What does "Project Numidia" aim to achieve in one sentence?

3. Geographic & Cultural Context

- o How might this project's location (Algeria) affect technical or content decisions?
- o Why is trilingual support mentioned as a requirement?

Phase 1B: Stakeholder Identification & Analysis

Instructions: Identify all stakeholder groups and complete the following table.

Stakeholder Group	Primary Goals	Pain Points (Current Problems)	Influence Level	Questions We Should Ask
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Academic Researchers				
Librarians/Archivists				
University Students				
General Public				
BNA Management				
[Your Group] - <i>Who else?</i>				

Critical Thinking Questions:

- Are there any stakeholders mentioned indirectly (e.g., UNESCO mentioned in the document)?
 - What conflicts might exist between stakeholder needs?
 - Which stakeholders have conflicting priorities regarding access vs. security?
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Task 2/ REQUIREMENT ELICITATION & CLARIFICATION

Phase 2A: Implicit vs. Explicit Requirements

Instructions: Requirements are stated explicitly ("must") or implied (between the lines). Categorize the requirements you find:

EXPLICIT REQUIREMENTS (Clearly stated):

- List 5 explicit functional requirements
- List 3 explicit non-functional requirements
- Example format: "The system **MUST** support Arabic, English, and French interfaces"

IMPLICIT REQUIREMENTS (Inferred from context):

- What must be true about the system for researchers to "compare manuscripts side-by-side"?
- What infrastructure is implied by "high-resolution images" and "responsive performance"?
- What compliance or legal requirements might apply to a national cultural institution?

AMBIGUOUS OR MISSING REQUIREMENTS: Create a list of 8-10 questions you would ask the client to clarify ambiguities. Examples:

- "You mention OCR initiative for full-text search. Is OCR accuracy guaranteed for all historical documents in Arabic?"
- "What does 'restricted and only viewable by authenticated researchers' mean exactly? How is a researcher authenticated?"
- "What is your expected user load? Are we building for 100 users or 100,000?"

Phase 2B: Requirement Classification Framework

Instructions: For EACH requirement identified, classify it using the following dimensions:

Requirement	Type	Category	Priority	Stakeholder	Status
Trilingual interface	Functional	User Interface	MUST	All Users	Explicit
Side-by-side manuscript view	Functional	Viewer Feature	SHOULD	Researchers	Explicit
High-resolution image support	Non-Functional	Performance	MUST	Researchers	Explicit
50-year data preservation	Non-Functional	Quality	MUST	Institution	Implicit

Legend:

- **Type:** Functional / Non-Functional / Constraint
- **Category:** User Interface, Search, Viewer, Authentication, Data Management, Performance, Security, Compliance, etc.
- **Priority:** MUST / SHOULD / COULD / WON'T (MoSCoW Method)
- **Status:** Explicit / Implicit / Ambiguous

Task 3/ FUNCTIONAL & NON-FUNCTIONAL REQUIREMENTS ANALYSIS

Phase 3A: Functional Requirements Deep Dive

Instructions: For each major feature area, expand the requirement into detailed components.

EXAMPLE: Public Search & Discovery

Stated requirement: "Advanced Search Engine: Search by metadata, full-text search within transcribed documents, ability to filter and sort results"

Your Turn - Expand this into detailed functional requirements:

1. The system SHALL provide a search interface that accepts free-text queries
2. The system SHALL support boolean operators (AND, OR, NOT) in search queries
3. The system SHALL allow filtering by: [List all filterable fields]
4. The system SHALL allow sorting results by: [List sort options]
5. The system SHALL display search results with: [What information is shown?]
6. The system SHALL highlight search terms in the result preview
7. The system SHALL support saved search queries for authenticated users
8. ... *continue identifying detailed functional requirements*

Now Analyze These Document Viewer Requirements:

- High-fidelity viewing of TIFF and JPEG2000 images
- Deep zoom, rotation, brightness/contrast controls
- Side-by-side comparison view
- Metadata displayed alongside document

Exercise: Break these into 10-12 specific, testable functional requirements using this format:

- "The system SHALL..."
- "The system MUST NOT..."
- "The system SHOULD..."

Phase 3B: Non-Functional Requirements Analysis

Instructions: Identify and specify non-functional requirements across multiple dimensions.

Performance:

- How fast should search results appear?
- What is acceptable image load time for high-resolution TIFF files?
- How many concurrent users must the system support?
- *Question to ask the client:* "What is your expected peak user load?"

Security:

- What level of authentication is required (username/password, OAuth, institutional login)?
- How should sensitive materials be access-controlled?

- What encryption standards should be used (in transit/at rest)?
- Who can upload new materials, and what approval workflow exists?

Availability & Reliability:

- What uptime percentage is required (99%, 99.9%)?
- What is the acceptable recovery time if the system fails?
- Must the system have backup/redundancy?

Usability & Accessibility:

- What accessibility standards must be met (WCAG 2.1 Level AA)?
- What languages must be fully supported?
- Should the system support mobile devices?
- *Question to client:* "What is your accessibility compliance requirement?"

Scalability:

- How much data (in TB/PB) must the system eventually manage?
- How should the system perform as the collection grows?

Maintainability:

- What is the expected lifespan of the digital assets (10 years, 50 years, indefinite)?
- What file formats must be supported long-term (TIFF, JPEG2000, PDF)?

Compliance:

- Are there UNESCO standards for digital preservation the system must meet?
- What data protection regulations apply (GDPR for EU researchers, local Algerian regulations)?

Create a table summarizing all identified NFRs:

NFR Category	Requirement	Measurable Criterion	Priority
Performance	Fast search response	< 2 seconds for 10,000 documents	MUST
Security	User authentication	OAuth 2.0 or institutional login	MUST
Availability	System uptime	99.5% annually	SHOULD

Task 4: USE CASE & USER STORY DEVELOPMENT

Phase 4A: Primary Use Cases

Instructions: Identify and document 5-6 critical use cases. Example format provided:

Use Case 1: Search Manuscripts by Period and Region

- **Actor(s):** Academic Researcher
- **Precondition:** User is on portal homepage; collection is indexed
- **Main Flow:**
 1. User enters search term "Ottoman period manuscripts"
 2. User filters by region: "Constantine"
 3. User sorts by date: "Newest First"
 4. System returns 23 matching manuscripts
 5. User clicks on first result
 6. System displays manuscript in high-resolution viewer
- **Alternate Flows:** Search returns no results; User applies date range filter
- **Postcondition:** Manuscript is displayed; user can proceed to annotation

Your Turn: Document 3 additional use cases covering:

1. [Researcher comparing two manuscripts]
2. [Archivist uploading new digitized items]
3. [Student discovering materials for coursework]

Phase 4B: User Stories with Acceptance Criteria

Instructions: Convert requirements into user stories using the format:

As a [stakeholder] **I want to** [action] **So that** [benefit]

Example:

- **As a** researcher, **I want to** save my search results to a personal collection, **so that** I can access them later without re-running the search.

Acceptance Criteria:

- User can click "Add to Collection" on any search result
- System prompts user to select or create a collection
- User can name a new collection
- Collection is saved to user profile
- User can retrieve saved collections from their dashboard

Your Assignment: Create 8-10 user stories (covering multiple stakeholders) with detailed acceptance criteria.

Task 5: CONSTRAINT ANALYSIS

Phase 5A: Technical & Organizational Constraints

Instructions: Identify constraints that limit design choices.

Constraint	Type	Impact	Questions
"System must be highly secure"	Security	Access controls required; encryption mandated	What is the threat model?
"Arabic and French must be identical"	Localization	Doubles UI/UX effort; language direction (RTL) issues	Are there other constraints on translations?
"Fast response time required"	Performance	CDN/caching strategy; image compression needed	What latency is acceptable?

Add 5-7 more constraints derived from the document.

Phase 5B: Risk Identification

Instructions: For each major constraint, identify potential risks:

Risk	Likelihood	Impact	Mitigation Strategy
OCR accuracy is poor for historical Arabic scripts	High	Full-text search becomes unreliable	Plan manual transcription for critical items; hybrid search strategy
Large file sizes slow down viewing experience	High	Users frustrated; page abandonment	Implement lazy loading, caching, CDN delivery

Task6/ FORMAL SRS OUTLINE

Instructions: Using everything you've learned in previous phases, create a formal Software Requirements Specification (SRS) document following the IEEE 830 standard.

Task 7/ QUESTIONS FOR STAKEHOLDER CLARIFICATION

Instructions: Prepare a formal list of clarification questions to send back to the client. Organize by category:

FUNCTIONAL CLARIFICATIONS:

1. What specific metadata fields should be searchable? (e.g., title, author, date, place, subject, material type, condition?)

2. Should researchers be able to download high-resolution images, or only view them online?
3. What citation formats beyond "APA, MLA, Chicago" are needed?

PERFORMANCE & SCALE:

1. What is your target user base size (concurrent users, registered users)?
2. What is the current size of your digitized collection, and what is the growth rate?
3. What are the technical limitations of your current infrastructure?

SECURITY & ACCESS CONTROL:

1. How do you define "authenticated researcher"? What is the authentication mechanism?
2. What percentage of your collection is restricted vs. publicly available?
3. How long should access logs be retained?

BUSINESS & ORGANIZATIONAL:

1. What is the project budget and timeline?
2. Will BNA host this system internally, or use cloud services?
3. Who will maintain the system long-term (in-house team, vendor support)?

DATA MANAGEMENT:

1. What is your long-term digital preservation strategy?
2. Do you currently have OCR-processed text for manuscripts, or is this a future initiative?
3. What formats are your digitized assets currently in?

Task 8/ Discussion Section

After completing the formal requirements analysis, consider how this process applies in real-world software development. Discuss the following in class:

Q1. Detailed analysis is costly. What business risks arise when doing such work before a client signs a contract? How does a paid *Discovery Phase* address these risks?

Q2. If you only had a short "Introduction and Vision," what collaborative methods (e.g., user story mapping, UX/UI workshops) and tools (e.g., Figma, Miro) would help turn it into clear requirements? Why is collaboration often more effective than requesting a formal document?

Q3. Agile teams don't produce full SRS documents every sprint. Does that make formal analysis irrelevant? Discuss how its rigor can still improve user stories, estimations, and backlog quality.